

Algebra – APTITUDE LEARNING NOTES

Complete concepts • formulas • shortcuts • solved examples • practice

1. Algebra Basics

Variable: a symbol whose value can vary. **Constant:** fixed value. **Coefficient:** numerical multiplier. **Term:** part separated by + or –.

Like terms have the same variables raised to the same powers: $3x$ and $-5x$ are like; $3x$ and $3x^2$ are unlike.

FORMULA: $a(b+c)=ab+ac$; $a(b-c)=ab-ac$.

2. Algebraic Identities

FORMULA: $(a+b)^2=a^2+2ab+b^2$.

FORMULA: $(a-b)^2=a^2-2ab+b^2$.

FORMULA: $a^2-b^2=(a-b)(a+b)$.

FORMULA: $(a+b)^3=a^3+3a^2b+3ab^2+b^3$.

FORMULA: $(a-b)^3=a^3-3a^2b+3ab^2-b^3$.

FORMULA: $a^3+b^3=(a+b)(a^2-ab+b^2)$.

FORMULA: $a^3-b^3=(a-b)(a^2+ab+b^2)$.

FORMULA: $a^3+b^3+c^3-3abc=(a+b+c)(a^2+b^2+c^2-ab-bc-ca)$.

3. Linear Equations

Standard form: $ax+b=c$, $a \neq 0$.

FORMULA: $ax+b=c \rightarrow x=(c-b)/a$.

Keep variable terms on one side and constants on the other; perform the same operation on both sides.

4. Simultaneous Equations

For $ax+by=c$ and $dx+ey=f$, use substitution or elimination.

Elimination is often fastest in aptitude: make one coefficient equal, add/subtract, then back-substitute.

5. Quadratic Equations

Standard form: $ax^2+bx+c=0$, $a \neq 0$.

FORMULA: $x=[-b \pm \sqrt{b^2-4ac}]/(2a)$.

Discriminant $D=b^2-4ac$: $D>0$ two distinct real roots; $D=0$ equal roots; $D<0$ no real roots.

FORMULA: Sum of roots $=-b/a$; product of roots $=c/a$.

6. Factorization

Take common factors first, then use identities. For x^2+bx+c , find two numbers with product c and sum b .

FORMULA: $x^2-(p+q)x+pq=(x-p)(x-q)$.

7. Exponents and Surds

FORMULA: $a^m \cdot a^n=a^{(m+n)}$; $a^m/a^n=a^{(m-n)}$; $(a^m)^n=a^{(mn)}$; $a^{(-n)}=1/a^n$.

FORMULA: $\sqrt{a} \cdot \sqrt{b} = \sqrt{ab}$.

Rationalization: $1/\sqrt{a} = \sqrt{a}/a$ for positive a .

8. Word Problems

Translate: 'twice a number' $\rightarrow 2x$; '5 more than x ' $\rightarrow x+5$; '3 less than twice x ' $\rightarrow 2x-3$.

Consecutive integers: $x, x+1, x+2$. Consecutive even/odd integers: $x, x+2, x+4$.

Solved Aptitude Examples

Q1. Solve $5x-9=31$.

Solution: $5x=40 \rightarrow x=8$.

Q2. If $x+y=12$ and $x-y=4$, find x,y .

Solution: Add: $2x=16 \rightarrow x=8, y=4$.

Q3. Factor $x^2-9x+20$.

Solution: $(x-4)(x-5)$.

Q4. Solve $x^2-7x+12=0$.

Solution: $(x-3)(x-4)=0 \rightarrow x=3 \text{ or } 4$.

Q5. If $a+b=10$ and $ab=21$, find a^2+b^2 .

Solution: $100-42=58$.

Q6. Simplify $(x^2-25)/(x-5)$.

Solution: $(x-5)(x+5)/(x-5)=x+5, x \neq 5$.

Q7. Three consecutive integers have sum 72.

Solution: $x+x+1+x+2=72 \rightarrow x=23 \rightarrow 23,24,25$.

Q8. Solve $2x/3+5=11$.

Solution: $2x/3=6 \rightarrow x=9$.

Q9. If $x+1/x=5$, find x^2+1/x^2 .

Solution: $25-2=23$.

Q10. Solve $2x^2-5x-3=0$.

Solution: $(2x+1)(x-3)=0 \rightarrow 3, -1/2$.

Practice Questions

1. Solve $7x-11=31$.

2. Solve $3x+5=26$.

3. If $x+y=18$ and $x-y=6$, find x and y .

4. Factor $x^2-11x+30$.

5. Solve $x^2-9x+20=0$.

6. Find a^2+b^2 if $a+b=14$ and $ab=40$.

7. Simplify $(x^2-16)/(x-4)$.

8. Find three consecutive integers with sum 90.

9. Solve $4x/5-3=9$.

10. If $x+1/x=4$, find x^2+1/x^2 .

11. Solve $3x^2-10x+3=0$.

12. Find the roots of $x^2-16=0$.

13. Simplify $(a+b)^2-(a-b)^2$.

14. If $2x+3y=19$ and $x+y=7$, find x,y .

15. A number and its square sum to 30. Find the positive number.

Answer Key

1) $6 \cdot 2$ 7 $\cdot 3$ $x=12, y=6 \cdot 4$ $(x-5)(x-6) \cdot 5$ $4, 5 \cdot 6$ $116 \cdot 7$ $x+4 \cdot 8$ $29, 30, 31 \cdot 9$ $15 \cdot 10$ $14 \cdot 11$ $3, 1/3 \cdot 12$ $\pm 4 \cdot 13$ $4ab \cdot 14$ $x=2, y=5 \cdot 15$ 5

Quick Revision Formula Sheet

- $(a+b)^2 = a^2 + 2ab + b^2$
- $(a-b)^2 = a^2 - 2ab + b^2$
- $a^2 - b^2 = (a-b)(a+b)$
- $a^3 \pm b^3$ identities
- Quadratic formula
- Sum/product of roots
- Exponent laws
- Consecutive integer forms

Exam strategy: Identify the pattern first, choose the shortest valid formula, solve step-by-step, and verify the final answer.